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THE GOVERNED RESEARCH LOOP

An Agentic Architecture for Quantum-Inspired and Machine-Learning FX Risk Research under Target-Zone Barriers

"The purpose of an autonomous research system is not to produce a winner. It is to organize a contest in which an attractive hypothesis is permitted to lose."

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Abstract

This paper explains the implementation of a governed research loop: an agentic architecture that converts a scientific question into a controlled sequence of hypothesis formation, experiment execution, evidence evaluation, memory update and termination. The architecture is illustrated through a reproducible laboratory for foreign-exchange risk under a target-zone barrier. The laboratory asks whether quantum-inspired tunneling representations, used strictly as nonlinear feature generators, improve the out-of-sample estimation of a future barrier breach when combined with classical machine-learning models.

The implementation separates four layers. Target-zone economics supplies the institutional and state-space structure. Quantum-inspired mappings translate barrier distance, market pressure and policy credibility into WKB-style, rectangular-barrier and resonant escape features. Machine learning estimates empirical relations using logistic regression, random forests, histogram gradient boosting and probability blends. The agentic layer governs which candidate families may be tested, preserves a locked chronological holdout, records an audit trail, enforces experiment and cycle budgets, and terminates under explicit rules.

The default synthetic experiment contains 2,730 usable observations and 208 ten-day forward breach labels. The design period contains 2,184 observations and 177 events; the untouched holdout contains 546 observations and 31 events. Across five walk-forward folds with a ten-day gap, the loop evaluates 47 candidates. A blend of a quantum-only logit and a quantum–machine-learning random forest minimizes the design-period decision loss, improving it by 8.41 percent relative to the best classical candidate. Yet the selected blend underperforms a historical-rate benchmark on the locked holdout: decision loss is 0.3585 versus 0.3189, Brier loss is 0.0620 versus 0.0541, and ROC AUC is 0.3414. The governance gate therefore returns *Rejected—Research Only*.

The negative deployment decision is the central result. It shows that a research loop should optimize neither novelty nor validation performance in isolation. It should preserve the institutional conditions under which a hypothesis can be falsified. The paper develops the formal architecture, explains the relationship among internal tools, plugins, MCP servers and APIs, documents the implemented experiment, interprets the results, and proposes an institutional extension consistent with contemporary model-risk and AI-risk governance.

Keywords: agentic research; autonomous scientific discovery; target zones; foreign-exchange risk; quantum-inspired finance; tunneling; machine learning; rare events; walk-forward validation; model risk; auditability.

JEL classification: C45; C52; C53; C55; F31; G17.

Contents

1 Introduction: From an Agentic Loop to a Research Institution

An agentic loop gives a model continuity. It receives a goal, observes a state, proposes an action, uses a tool, receives the consequence, updates memory and decides whether to continue. That architecture is sufficient to transform a one-shot inference into a path-dependent process [Reynoso, 2026]. It is not, however, sufficient to transform that process into research.

Research imposes a more demanding constitution. A system may repeatedly fit models and still fail to perform science. It may explore a large parameter grid, select the smallest validation loss and generate a polished report while having no protected evidence, no falsifiable claim, no economic benchmark and no mechanism for preserving negative findings. It can be highly automated and epistemically weak.

The **governed research loop** developed here is an agentic loop specialized for scientific inquiry. Its state contains not only a goal and tool results but also hypotheses, admissibility conditions, datasets, benchmarks, experiment budgets, validation records, a protected holdout and an audit history. Its policy gate does not merely ask whether a tool call is authorized. It asks whether the proposed experiment is scientifically admissible. Its stopping rule does not merely detect task completion. It distinguishes selection, validation, rejection, inconclusive evidence, budget exhaustion and safe failure.

This distinction is becoming important as large-language-model systems increasingly participate in the research lifecycle. Recent autonomous-science systems generate ideas, implement experiments, analyze results and draft papers [Lu et al., 2024]. Automated machine learning already frames model choice and hyperparameter selection as a joint optimization problem [Feurer et al., 2015], while Bayesian optimization allocates scarce experiment budgets [Snoek et al., 2012]. These developments make research faster, but speed intensifies a familiar statistical hazard: the more adaptively the researcher searches, the easier it becomes to discover a model that explains the validation environment by accident.

The application is foreign-exchange risk when the exchange rate is constrained by a lower and upper target-zone barrier. This setting is unusually suitable for the argument. Target-zone economics provides a mature theory of nonlinear dynamics, credible defense, smooth pasting, intervention and realignment [Krugman, 1988, Flood and Garber, 1991, Svensson, 1992]. The defended band also invites a quantum-tunneling analogy: a state appears confined within a range but may escape when the effective barrier becomes sufficiently thin, low or penetrable. That analogy can be mathematically productive, but it can also be seductive. A governed loop is therefore asked to determine whether the nonlinear representation earns predictive value, not to confirm that the market is quantum.

Research question

> When an exchange rate is bounded by a target-zone barrier, does an explicit quantum-inspired representation of barrier geometry improve the governed, out-of-sample estimation of a breach event after conventional economic and machine-learning alternatives are admitted to the same contest?

The contribution is architectural rather than a claim of empirical superiority. The paper explains a working Colab implementation in which the research process itself becomes executable. The reasoner selects the next phase; a policy gate protects methodological constraints; tools conduct time-aware backtests; observations are metric tables and out-of-fold probability vectors; memory is a leaderboard plus an audit log; and termination occurs only after the champion has been frozen or a limit has been reached.

The main result is deliberately uncomfortable. The quantum–ML blend wins during model selection but fails on the untouched holdout. A conventional workflow might quietly retune, redefine the event or replace the evaluation period. The governed loop refuses. It records the failure and rejects deployment.

> *"An autonomous researcher becomes scientifically useful when it can preserve a result that defeats its own preferred hypothesis."*

2 What "Research Loop" Means

2.1 The logical structure

Let the research state at cycle t be

$$s_t = \big(g, \mathcal{D}_t, \mathcal{H}_t, \mathcal{E}_t, \mathcal{M}_t, \rho, b_t\big),$$

where g is the research goal, $\mathcal{D}_t$ is the admissible data state, $\mathcal{H}_t$ is the current hypothesis set, $\mathcal{E}_t$ is accumulated experimental evidence, $\mathcal{M}_t$ is research memory, ρ is the methodological and institutional policy, and b_t is the remaining budget.

A reasoner proposes an action

$$a_t \sim \pi(a \mid s_t).$$

The proposal may be to establish a benchmark, generate a feature family, fit a model, form an ensemble, inspect an error, request new data, freeze a specification, unlock a protected dataset, stop or escalate. The methodological gate then applies

$$\widehat{a}_t = G(a_t, \rho, s_t).$$

The gate may authorize, reject, defer or modify the proposal. If authorized, an executor returns an observation:

$$o_t = E(\widehat{a}_t; \mathcal{D}_t).$$

The state update is

$$s_{t+1} = U(s_t, \widehat{a}_t, o_t),$$

followed by a termination function

$$\tau(s_{t+1}) \in \{\text{continue}, \text{freeze}, \text{validate}, \text{reject}, \text{inconclusive}, \text{fail safely}\}.$$

This is not ordinary hyperparameter optimization. The action space includes changes in the *research program*, not only changes in coefficients. Nor is it unrestricted autonomous science. The policy ρ fixes what evidence is available, which tools may be invoked, which transformations are permissible and when protected information may be revealed.

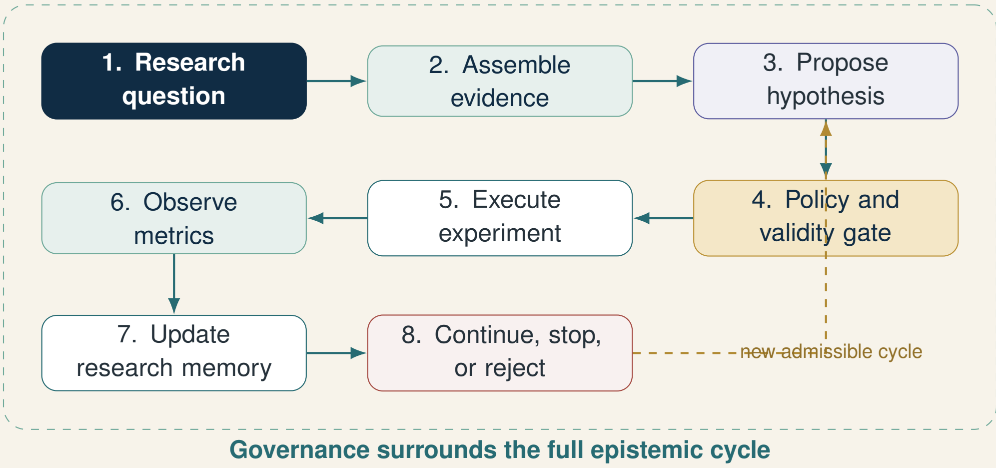

Figure 1: The agentic loop translated into a governed research loop.

2.2 The epistemic minimum

A research loop requires more than the six logical ingredients of a generic agent. Its minimum viable constitution contains ten elements:

1. **A falsifiable question.** There must be a possible result under which the proposed representation is rejected or judged unnecessary.
2. **A declared estimand.** Here it is the probability that either barrier is crossed at least once during the next H business days.
3. **An admissible candidate space.** Classical, quantum-only, hybrid and ensemble families must be specified before protected evaluation.
4. **Benchmarks.** A historical event-rate predictor and classical ML models prevent novelty from competing only against weak alternatives.
5. **An executable experiment tool.** Each candidate must pass through the same time-aware fitting and scoring procedure.
6. **Comparable evidence.** Out-of-fold probability vectors, not training scores, form the basis of selection.
7. **Research memory.** The leaderboard and audit log preserve what was tested, in what order, with what result.
8. **Protected evidence.** A locked final holdout is unavailable to adaptive search.
9. **Budgets and stopping rules.** Cycles, candidate count and completion conditions are explicit.
10. **A rejection path.** The system must be able to preserve a negative result without rewriting the experiment.

2.3 Deterministic agency is still agency

The implemented reasoner is deterministic. It moves through five phases: baseline, quantum search, hybrid search, blend search and selection. This choice is intentional. A large language model could propose novel candidates or interpret failure modes, but reproducibility is more important than linguistic flexibility in the base experiment.

Agency does not require stochastic prose generation. It requires goal-directed choice, feedback, state and controlled continuation. A deterministic research policy can be agentic because later actions depend on earlier evidence: the best quantum configuration selected in phase two determines the hybrid feature map tested in phase three; the best classical, quantum and hybrid models determine the ensembles constructed in phase four; and the entire accumulated leaderboard determines the frozen champion in phase five.

This architecture also creates a clean extension point. An LLM may later propose an additional candidate, provided that the proposal is transformed into a typed experiment specification and passes the same gate. The LLM does not receive authority to inspect the holdout, redefine the target after failure or exempt its preferred hypothesis from a benchmark.

3 The Economic Object: Risk inside a Target Zone

3.1 Defended bands are not ordinary ranges

In the canonical target-zone model, expectations that monetary authorities will defend a band affect exchange-rate behavior before the boundary is reached. Credible defense therefore produces nonlinear dynamics and the familiar "honeymoon effect" [Krugman, 1988, Svensson, 1992]. The exchange rate is not merely truncated by an arbitrary support and resistance line; its law of motion incorporates beliefs about future policy.

Finite intervention capacity changes the problem. Linking target zones to speculative-attack models makes reserve exhaustion, discrete intervention and collapse relevant [Flood and Garber, 1991]. Realignment risk further weakens the idealized reflecting-boundary interpretation [Bertola and Caballero, 1992]. Empirically, target-zone exchange rates can exhibit both realignment jumps and jumps that remain within the band [Bekaert and Gray, 1998]. A useful risk model must therefore represent at least three phenomena:

- smooth movement inside the band;
- endogenous pressure near a defended boundary; and
- discontinuous or rare escape when the regime loses credibility or is realigned.

The laboratory uses synthetic data because the purpose is to expose the full mechanism without credentials, unresolved band definitions or historical-policy disputes. It does not claim that the generated series is an official target zone, nor that a rolling statistical band in public spot data is equivalent to a policy commitment.

3.2 The estimand is path-dependent

Let S_t be the exchange rate and L_t, U_t the contemporaneous lower and upper barriers. For a horizon H, the event label is

$$y_t^{(H)} = \mathbf{1}\{\exists j \in \{1, \ldots, H\} : S_{t+j} < L_{t+j} \ \vee \ S_{t+j} > U_{t+j}\}.$$

The label equals one if either barrier is crossed on *any* of the next H days. It is not a terminal-value indicator. A path that breaches on day three and returns inside the zone by day ten remains an event. This definition is appropriate for risk management because barrier contact can trigger intervention, option payoffs, stop-loss rules, collateral effects or governance escalation even if the period-end rate looks benign.

All predictors use information available at t. Since neighboring labels share future observations, the walk-forward validation gap is set equal to H. With $H = 10$, ten samples are excluded between the end of each training set and the beginning of its validation fold.

3.3 Classical state variables

The feature factory creates twenty barrier-aware variables. They include returns and momentum over several horizons, rolling volatility, relative position inside the band, distance to the nearest barrier, band width, recent boundary pressure, escape history and proxies for credibility or intervention intensity.

No single variable is declared structural. The objective is to offer conventional learners an economically interpretable state space before adding a tunneling transformation. This ordering matters. If quantum-inspired features are compared only against the raw spot rate, any gain may simply reflect the advantage of encoding distance to the boundary. The classical layer therefore already contains barrier geometry in transparent form.

4 Quantum-Inspired Tunneling as a Testable Representation

4.1 Analogy, simulation and quantum computation

Three ideas must be separated.

1. **Quantum mechanics** is a physical theory with states, observables, operators and a specified dynamical law.
2. **Quantum computing** uses quantum information processing and algorithms whose implementation depends on quantum hardware or simulation [Biamonte et al., 2017, Cerezo et al., 2021].
3. **Quantum-inspired modeling** borrows a mathematical form or geometric intuition and evaluates it as a classical computational representation.

The notebook belongs entirely to the third category. It uses NumPy, pandas and scikit-learn on a classical runtime. It does not prepare qubits, construct a variational circuit, execute a quantum kernel or claim quantum advantage. "Effective $\hbar$" and related parameters are dimensionless tuning constants. They are not measurements of physical constants in a currency market.

This boundary is not a semantic precaution. It determines the standard of evidence. A quantum algorithm would require a hardware and complexity argument. A quantum-inspired feature requires evidence that its nonlinear map adds out-of-sample information relative to transparent alternatives. The latter is the question asked here.

4.2 The WKB representation

In one-dimensional quantum mechanics, the WKB approximation yields an exponentially small transmission probability through a classically forbidden region [Griffiths and Schroeter, 2018]. The laboratory maps market variables into dimensionless analogues:

$$\text{distance to barrier} \mapsto a_t, \qquad \text{pressure or effective energy} \mapsto E_t, \qquad \text{defense or credibility} \mapsto V_t.$$

The feature is

$$T_t^{\mathrm{WKB}} = \exp\left[-2\gamma a_t \sqrt{\frac{(V_t - E_t)_+}{h_{\mathrm{eff}}}}\right].$$

The transformation captures an economically plausible asymmetry. Escape likelihood changes slowly when the state is deep inside a credible band, but may change rapidly when distance narrows, pressure rises or the effective defense falls. The exponential makes interactions among those inputs explicit rather than asking every learner to discover them from finite rare-event data.

4.3 Rectangular and resonant representations

The second family uses a finite rectangular-barrier transmission analogue with a hyperbolic-sine penalty. The third introduces a two-sided resonant term that permits location-dependent amplification between lower and upper defenses. Each parameterization produces six quantum-inspired variables.

The search grid contains twenty-four configurations defined by barrier scale, action scale, effective h and resonance frequency. This is a deliberately small, governed grid. It does not attempt to estimate a physical Hamiltonian. It asks whether one of several transparent nonlinear encodings provides useful features.

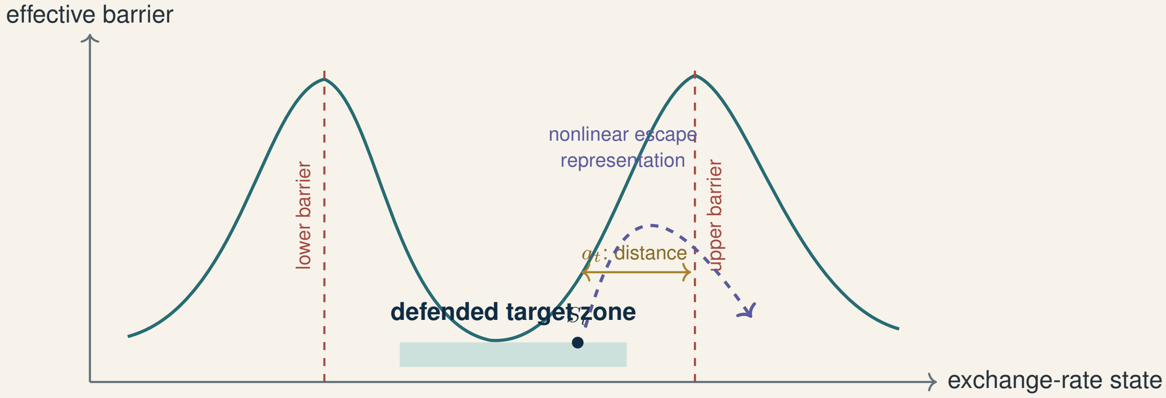

Figure 2: Target-zone geometry as a feature-generating analogy. The curve is not a claim that foreign exchange obeys a physical Hamiltonian.

The correct null hypothesis

The null is not that quantum tunneling is “false” in a metaphysical sense. The operational null is that the proposed tunneling transformations add no decision-relevant out-of-sample information after barrier-aware classical variables and conventional nonlinear learners are available.

4.4 Why the mapping may fail

The representation can fail for substantive reasons. Policy credibility may be discontinuous rather than smoothly encoded. Reserve adequacy, offshore derivatives, intervention announcements and political realignment may dominate price-based signals. A single effective barrier can confuse lower- and upper-zone dynamics. The synthetic process may not resemble any historical regime. Finally, a flexible ML model may already approximate the useful nonlinearity from classical variables, rendering the handcrafted map redundant.

The research loop treats each failure as permissible. A classical winner would imply that the quantum map is unnecessary for the dataset. A quantum-only winner would invite scrutiny of omitted classical variables. A hybrid winner would suggest an interaction worth further study. A blend winner would suggest partially distinct errors. None of these outcomes would establish a physical quantum mechanism.

5 The Machine-Learning Contest

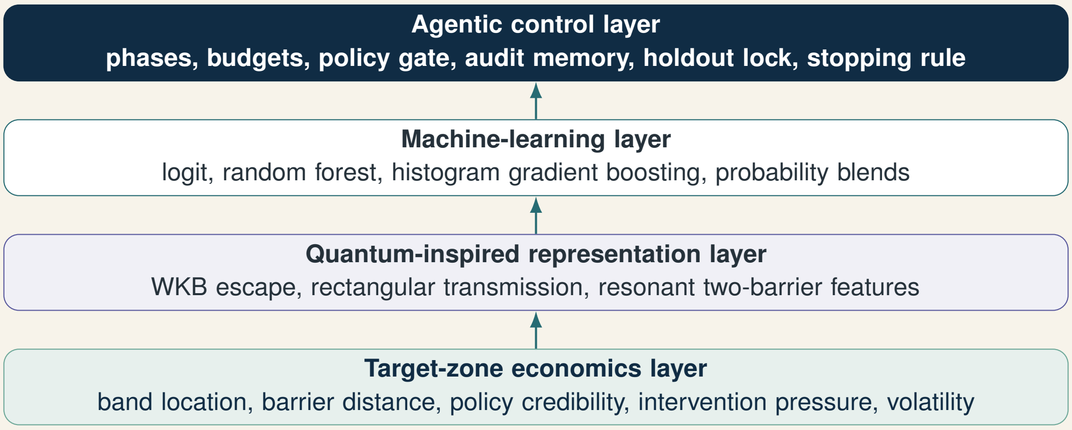

Figure 3: Four layers of the laboratory. The control layer governs the search; it does not validate the economic or physical interpretation by assertion.

5.1 Candidate families

The laboratory constructs four classes:

1. **Classical**: barrier-aware economic variables passed to logistic regression, random forest and histogram gradient boosting.
2. **Quantum-only**: the six tunneling features passed to logistic regression across twenty-four parameter configurations.
3. **Hybrid**: classical and quantum-inspired variables jointly passed to logit, random forest and histogram gradient boosting, using the best quantum map from the preceding phase.
4. **Probability blends**: convex combinations of the best quantum and hybrid forecasts, and of the best classical and hybrid forecasts.

Logistic regression supplies a regularized, interpretable benchmark. Random forests model nonlinear interactions through randomized tree ensembles [Breiman, 2001]. Histogram gradient boosting provides a different nonlinear approximation based on additive tree construction, related to the general gradient-boosting framework [Friedman, 2001]. The family design is not intended to exhaust modern ML. It is intended to create a compact contest in which the source of any gain can be interpreted.

5.2 Why probability quality matters

Barrier breach is a rare binary event. Accuracy is nearly useless because a classifier that always predicts “no breach” can appear successful. The laboratory therefore scores probabilities using:

- Brier loss, a proper squared-probability score [Brier, 1950];
- log loss, which sharply penalizes confident errors;
- ROC AUC for ranking discrimination;
- precision-recall AUC, which is informative under class imbalance [Davis and Goadrich, 2006, Saito and Rehmsmeier, 2015];
- expected calibration error;
- recall captured in the top probability decile.

The governed selection loss is

$$\begin{aligned}\mathcal{L} &= 0.40\,\mathrm{Brier} + 0.15\,\mathrm{LogLoss} + 0.15(1 - \mathrm{ROC\ AUC}) \\ &\quad + 0.15(1 - \mathrm{PR\ AUC}) + 0.10\,\mathrm{ECE} + 0.05(1 - \mathrm{Recall}_{10\%}).\end{aligned} \tag{1}$$

Lower is better. The weights are a declared research preference, not a universal optimum. They prioritize calibrated probabilities while retaining discrimination and rare-event concentration. A production institution would derive weights from the economic loss function of the exposure, intervention policy, capital charge or hedging decision.

5.3 Time-aware evidence

Ordinary random cross-validation would allow models to train on observations later than their validation targets. The implementation uses expanding-window `TimeSeriesSplit`. Successive training sets are supersets of earlier sets, while the validation fold always lies in the future [scikit-learn developers, 2026]. A ten-day gap separates each training window from its validation fold.

Every candidate returns both fold-level metrics and a single out-of-fold probability vector. The latter permits like-for-like scoring and construction of probability blends without using fitted values. Model selection therefore uses simulated historical deployment, not in-sample fit.

6 Implementation of the Executable Research Loop

6.1 Research configuration

The notebook exposes a configuration object rather than scattering constants through code. Its default values include:

Control	Default	Research meaning
Data mode	Synthetic	Reproducible target-zone laboratory without credentials
Observations	2,800	Pre-feature-engineering sample
Band center / half-width	18.00 / 0.80	Synthetic defended interval
Breach horizon	10 days	Path-dependent forecast event
Holdout fraction	20%	Protected final chronological period
Walk-forward folds	5	Expanding historical validation
Gap	10 days	Matches the forward-label horizon
Maximum cycles	5	Bounds agentic continuation
Maximum candidates	80	Prevents uncontrolled search
Minimum events	25	Blocks an inadequately populated design sample
Seed	367	Reproducible stochastic components

The same notebook supports a CSV contract with four columns: `Date`, `FX`, `LowerBarrier` and `UpperBarrier`. It can also retrieve a public spot series, but public spot data do not supply an official policy band. If rolling statistical bands are used, the notebook labels them as a laboratory approximation.

6.2 The five autonomous cycles

The reasoner maps phase to action:

Cycle	Tool	Purpose and state transition
1. Baseline	`evaluate_baselines`	Fit the historical-rate benchmark and three classical learners; move to quantum search.
2. Quantum	`search_quantum_models`	Evaluate twenty-four WKB/rectangular/resonant maps with logit; retain the best map; move to hybrid search.
3. Hybrid	`evaluate_hybrids`	Combine the best quantum map with classical variables in three learners; move to blend search.
4. Blend	`optimize_blends`	Search convex probability combinations using out-of-fold predictions; move to selection.
5. Select	`select_champion`	Freeze the lowest-loss specification, complete adaptive search and authorize one holdout evaluation.

Each action is proposed, checked against an allow-list, executed and recorded. The gate rejects an action if the cycle budget is exhausted, the candidate budget is exhausted, the tool is not allowed or the holdout has been unlocked prematurely.

6.3 The holdout as a governed resource

The holdout is not simply a dataframe whose use is discouraged in prose. Its lock is a state variable. Adaptive phases are authorized only while `holdout_locked=True`. The selection tool freezes the champion and changes the state once. No model family, feature set, blend weight or hyperparameter may be changed in response to the resulting metrics.

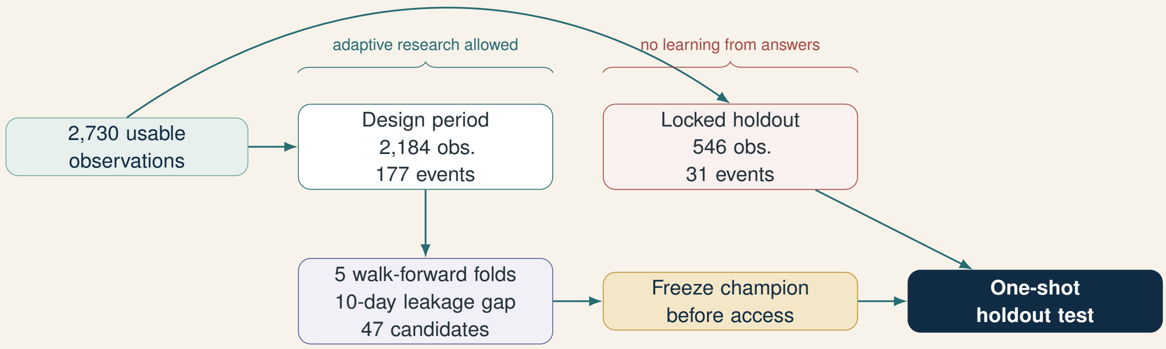

Figure 4: The holdout is a governed resource, not another dataset available to the search loop.

This makes the boundary institutional. The system can execute many experiments on the design period, but it does not possess the authority to learn from the final answer key. If the holdout result is poor, the admissible next action is to document rejection and begin a new research program with a genuinely new protected period.

6.4 Research memory and auditability

Two memories are maintained. The **epistemic memory** is the results list, including specifications, metrics, out-of-fold predictions and fold-level evidence. The **institutional memory** is the audit log, containing timestamp, cycle, phase, event and relevant details.

At the end of the run, a manifest records:

- configuration and random seed;
- data fingerprint;

- design and holdout periods;
- confirmation that the holdout remained locked;
- number of cycles and candidates;
- best quantum configuration and champion specification;
- holdout and benchmark metrics;
- deployment status; and
- scientific and operational limitations.

The notebook can export the leaderboard, holdout report, stress scenarios, audit log and experiment manifest. The report is therefore downstream of evidence rather than a substitute for it.

7 Plugins, MCP, APIs and Internal Research Tools

7.1 The notebook implementation

The base laboratory uses internal Python functions as tools. Data generation, feature construction, candidate evaluation, blend optimization, risk conversion and audit export all occur within the Colab runtime. The reasoner does not need a network protocol to call these functions because they share one process and one state.

This is the simplest form of tool-augmented research:

$$\text{reasoner} \rightarrow \text{typed local tool} \rightarrow \text{result object} \rightarrow \text{research memory.}$$

An internal tool should still have a contract: input specification, output schema, side effects, possible exceptions and reproducibility requirements. Locality removes network complexity; it does not remove the need for governance.

7.2 Extension to external resources

A production research loop may need central-bank releases, market data, reserve series, news, policy documents, databases, code repositories, cloud quantum simulators or enterprise model inventories. Four interface concepts then become relevant:

1. **API**: a service-specific transaction boundary. It defines endpoints, authentication, parameters and responses. A market-data library may wrap several APIs.
2. **MCP server**: a standardized capability boundary through which an AI host can discover and invoke tools or access resources. The MCP server may call an API internally, query a database or execute local code.
3. **Plugin**: a deployable package that can combine tools, skills, MCP servers, metadata and policy configuration for a host environment.
4. **Skill**: procedural knowledge describing how the research task should be performed, validated and documented. A skill can instruct the loop to use a tool but does not itself provide credentials or external access.

Layer	What it contributes	FX research example	Governance question
Skill	Method and quality contract	Define the breach label, gap and holdout rules	Is the method approved and versioned?
Tool	Executable capability	Run a walk-forward backtest	Are inputs and outputs validated?
MCP	Discovery and invocation protocol	Expose market-data and simulator tools	Which tools are visible and callable?
API	Underlying service transaction	Request historical FX or reserve data	Which account, scope and rate limit apply?
Plugin	Packaged integration	Install a complete FX-research capability	Who approved the package and dependencies?

The research loop remains the controller. MCP does not supply scientific judgment; an API does not preserve the holdout; a plugin does not create falsifiability. Those properties belong to the research constitution and orchestrator.

7.3 A safe external-data sequence

Suppose the loop proposes adding an intervention series. A safe path is:

1. the reasoner emits a typed request for a named series and period;
2. the policy gate checks provenance, licensing, credentials and whether acquisition occurs before the champion is frozen;
3. an MCP-exposed data tool validates the request and calls the provider API;
4. the response is normalized, fingerprinted and stored with provenance;
5. a data-quality tool checks missingness, revisions and temporal availability;
6. the research state records a new data version and begins a new experiment identifier.

The last step is essential. Adding a new variable after seeing the failed holdout does not repair the old experiment. It defines a new experiment that requires new protected evidence.

8 Experimental Results

8.1 Data and search scale

After feature construction and forward-label alignment, the synthetic laboratory contains 2,730 usable observations from 25 January 2016 to 10 July 2026. There are 208 ten-day forward breach labels, an event rate of 7.62 percent. The design period contains 2,184 observations and 177 events; the 20 percent locked holdout contains 546 observations and 31 events.

The loop completes five cycles and evaluates 47 candidates, below its budget of 80. It chooses the quantum configuration

$$(V, \gamma, h_{\text{eff}}, r) = (0.75, 2.5, 0.65, 1.0),$$

and selects a probability blend with weight $\alpha = 0.15$ on the hybrid random forest and 0.85 on the quantum-only logit.

8.2 Design-period selection

On walk-forward validation, the best classical candidate has decision loss 0.3661. The best quantum-only candidate records 0.3400, an improvement of 7.13 percent. The best hybrid alone records 0.3691 and therefore does not improve on the classical winner. The selected quantum–hybrid blend records 0.3353, an improvement of 8.41 percent relative to the best classical candidate.

Design-period candidate	Decision loss	vs. classical	Meaning
Best classical	0.3661	0.00%	Conventional reference
Best quantum-only	0.3400	+7.13%	Compact nonlinear map appears useful
Best hybrid	0.3691	−0.80%	Added features do not help this learner
Selected blend	0.3353	+8.41%	Diversified design-period errors

Had the analysis ended here, the result would have appeared supportive. The loop would have discovered a quantum-inspired specification, combined it with ML and reported a material reduction in the declared loss.

8.3 Locked-holdout result

The final period reverses the conclusion.

Holdout model	Brier	Log loss	ROC AUC	Decision loss
Selected quantum–ML blend	0.0620	0.2569	0.3414	0.3585
Historical-rate benchmark	0.0541	0.2224	0.5000	0.3189

The selected model is worse on every metric shown. Its precision-recall AUC is 0.0527 versus 0.0568 for the benchmark. Expected calibration error is 0.0593 versus 0.0243. Recall in the top probability decile is 0.0323 versus 0.1007. The deployment gate requires lower decision loss, lower Brier loss and ROC AUC above 0.50. All three conditions fail.

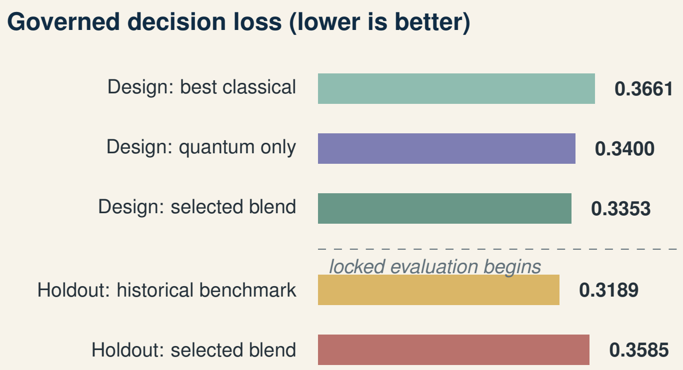

Figure 5: Selection success did not survive the untouched period. The loop therefore rejected deployment.

REJECTED—RESEARCH ONLY

This is not evidence that every tunneling representation is useless. It is evidence that the selected mapping and search design did not generalize to the protected synthetic period. The distinction is the essence of disciplined inference.

8.4 Severity, risk card and stresses

The notebook supplements event probability with conditional return quantiles estimated by gradient boosting. The nominal 98 percent interval attains only 95.24 percent empirical coverage on the holdout, another warning against deployment. The breach-conditioned downside expected-shortfall log return in the design sample is -7.572 percent.

At the final state, the selected model reports a 9.54 percent probability of any barrier breach during the next ten business days. For the illustrative linear exposure, the approximate 99 percent VaR is 272,370 monetary units and empirical breach-conditioned downside expected shortfall is 757,222. These values remain labeled as research outputs because the deployment gate failed and because nonlinear portfolios require full revaluation.

The governed stress layer produces:

Scenario	Predicted breach probability
Baseline	9.54%
Volatility $+50\%$	10.95%
Band narrows 20%	16.07%
Credibility shock	11.87%
Combined shock	21.65%

These stresses are diagnostic transformations, not forecasts of policy. Their value is to test qualitative monotonicity and sensitivity. A model that assigns lower escape risk after simultaneous volatility, credibility and band-width shocks would require immediate investigation.

9 Interpretation: What the Loop Learned

9.1 The winner was conditional on the research environment

"Optimal" always means optimal relative to a candidate set, data partition, metric, search budget and governance policy. The selected blend was optimal on the design-period loss among 47 admissible candidates. It was not optimal on the untouched environment and therefore was not validated for use.

This prevents a common category error. Model selection answers:

$$\widehat{m} = \arg\min_{m \in \mathcal{M}} \widehat{\mathcal{L}}_{\text{design}}(m).$$

Validation asks whether the frozen $\widehat{m}$ satisfies ex ante conditions on new data:

$$\text{Approve}(\widehat{m}) = \mathbf{1}\left\{\mathcal{L}_{\text{holdout}}(\widehat{m}) < \mathcal{L}_{\text{holdout}}(m_0),\ \text{Brier}_{\widehat{m}} < \text{Brier}_{m_0},\ \text{AUC}_{\widehat{m}} > 0.5\right\}.$$

The first operation must produce a winner. The second may produce a rejection.

9.2 Adaptive overfitting can occur without a neural network

The loop does not use an enormous model family, yet it still adapts. It chooses the best of twenty-four quantum maps, uses that choice to define hybrid candidates, and then searches blend weights. Each stage conditions on earlier design evidence. This is scientifically reasonable, but it increases the risk that the final specification captures peculiarities of the design period.

The locked holdout is therefore not an optional final chart. It estimates the performance of the *entire adaptive procedure*, not merely the fitted champion. If the same holdout were used repeatedly, it would become another training resource and cease to perform that function.

9.3 The negative result strengthens the architecture

The failed holdout demonstrates four properties.

1. **The loop is hypothesis-neutral.** It allows classical, quantum, hybrid or blend outcomes.
2. **The policy gate has causal force.** It prevents a post-failure change in the model.
3. **The audit record preserves embarrassment.** The design-period improvement and holdout deterioration remain in the same manifest.
4. **Termination is meaningful.** "Complete" describes completion of the research protocol, not success of the model.

This is a more important demonstration of autonomous research than a favorable synthetic backtest would have been. A system that always returns a positive result is not an autonomous scientist. It is a publication generator.

10 Institutional Governance

10.1 Two concentric governance scopes

The laboratory contains both quantitative models and an agentic control system. They should not be governed as though they were identical.

The quantitative risk models fall within the familiar concerns of model development, validation, outcomes analysis, monitoring, inventory and documentation. The 2026 U.S. interagency revised model-risk guidance emphasizes a risk-based approach, effective challenge, conceptual soundness, outcomes analysis and governance across the model lifecycle [Board of Governors of the Federal Reserve System et al., 2026]. The guidance explicitly states, however, that generative and agentic AI models are outside its scope. That exclusion does not make the orchestrator ungoverned. It means a second governance layer is required.

The NIST AI Risk Management Framework organizes AI risk activity around GOVERN, MAP, MEASURE and MANAGE and treats governance as cross-cutting throughout the lifecycle [National Institute of Standards and Technology, 2023]. The research loop naturally maps to this structure:

- **GOVERN**: define authority, holdout custody, candidate budgets, documentation and escalation;
- **MAP**: identify the FX use case, exposure, target-zone regime, affected decisions and failure consequences;
- **MEASURE**: conduct walk-forward tests, calibration analysis, stress testing and independent challenge;
- **MANAGE**: reject, restrict, redevelop, monitor or retire the model and its autonomous workflow.

10.2 Separation of duties

For consequential deployment, one agent or team should not simultaneously own hypothesis generation, holdout custody, validation criteria and release authority. A robust topology includes:

1. a research agent that proposes and executes admissible experiments;
2. a data custodian that controls protected datasets and provenance;
3. a validation agent or independent human function that tests conceptual soundness and outcomes;
4. a governance gate that applies predeclared thresholds;
5. an accountable human owner who accepts residual risk and approves use.

The agents may be software roles, organizational roles or both. The separation matters more than the anthropomorphic labels.

10.3 Deployment is not the next notebook cell

Before institutional use, the laboratory would require at least:

- historically documented target-zone episodes and realignment dates;
- reserve, rate, forward, option-implied, intervention and policy-announcement variables;
- temporal availability metadata that prevents revised data from entering earlier forecasts;
- exposure-specific P&L and full revaluation for nonlinear instruments;
- nested or rolling external validation across multiple regimes;
- uncertainty intervals, drift measures and challenger models;
- adverse testing for data gaps, regime changes, API failure and duplicated actions;
- independent validation and effective challenge;
- access controls, model inventory, versioning, incident response and retirement criteria.

The current result is educational and architectural. It is not investment advice, a production trading system or evidence of live-market performance.

11 Limitations and Research Agenda

11.1 Synthetic evidence

Synthetic data make the full loop reproducible, but they also encode the researcher's assumptions. A model can appear successful by recovering the generator's structure. Conversely, a model can fail because the holdout generator changes in a way that is unlike a real regime. The results demonstrate the governance mechanism; they do not estimate the frequency or severity of actual target-zone collapse.

11.2 Single protected period

One holdout gives one realization of generalization performance. It is stronger than reusing validation data but weaker than evaluation across multiple historical regimes, countries and policy frameworks. A mature study would use nested time-series validation and preserve at least one external episode untouched by model development.

11.3 Metric weights

The decision loss is transparent but judgmental. A treasury concerned with hedging cost may place more weight on calibration. A central bank concerned with missed attacks may emphasize tail recall. A trading desk may require economic utility net of transaction costs. Sensitivity to the loss weights should be studied in the design period and frozen before final evaluation.

11.4 Causal and institutional variables

Price-derived variables cannot fully represent policy credibility. A future research program should incorporate central-bank communication, reserve adequacy, fiscal conditions, offshore positioning, political events and market microstructure. Those additions should be dated by information availability and introduced through a new governed experiment.

11.5 Quantum representation

The three tunneling maps are hypotheses, not a complete theory. Their dimensionless mappings may be non-identifiable, and multiple parameter combinations may generate similar features. A stronger program would:

- derive barrier geometry from explicit economic microfoundations;
- test invariance across quotation conventions and band rescaling;
- compare against splines, kernels, neural networks and monotone boosting;
- use ablation tests to isolate each quantum-inspired feature;
- examine whether the mapping contributes to calibration, ranking or stress coherence;
- separate representational advantage from algorithmic or quantum-computational advantage.

11.6 Autonomous hypothesis formation

The deterministic reasoner is a strength for reproducibility and a limitation for discovery. An LLM extension could propose new economic variables, alternative barrier forms or diagnostic tests. It should do so within typed schemas, bounded tool access and a versioned protocol. Novelty generation may be autonomous; evidence custody must remain constrained.

12 Conclusion

The agentic loop becomes a research loop when its repeated actions are governed by an epistemic constitution. The system must know not only what to do next but what it is forbidden to learn from, which comparisons are admissible, what evidence is protected, how a negative result is recorded and when continuation would invalidate the experiment.

The FX target-zone laboratory implements this principle in a concrete sequence. Target-zone economics defines the bounded process. Quantum-inspired tunneling equations create nonlinear escape features. Machine-learning models estimate interactions and probability forecasts. The agentic layer sequences the experiment, enforces budgets, locks the holdout, preserves research memory and applies a stopping rule.

The loop discovers an apparently attractive result: a quantum–ML probability blend reduces design-period decision loss by 8.41 percent relative to the best classical candidate. The untouched period then rejects it. The selected blend has worse Brier loss, worse log loss, worse discrimination and worse composite decision loss than a historical-rate benchmark. The system ends successfully as an experiment and unsuccessfully as a model.

That distinction is the paper's central lesson. Autonomous research should not be evaluated by how often it finds something novel. It should be evaluated by whether it preserves the conditions under which novelty can be defeated by evidence.

Final proposition

The deepest contribution of the research loop is not automated optimization. It is automated methodological discipline: the capacity to generate hypotheses rapidly while making the right to validate, approve and act increasingly difficult.

A Pseudocode of the Implemented Loop

```
state = AgentState(
   goal="Select a governed FX barrier-risk model",
   phase="baseline",
   holdout_locked=True,
   candidate_budget=80,
   cycle_budget=5,
)

while state.outcome is None:
   proposal = reasoner(state)
   allowed, reason = policy_gate(state, proposal)

   audit(state, "proposal", proposal=proposal, gate=reason)

   if not allowed:
      state.outcome = "fail safely"
      break

   observation = execute_tool(
      proposal,
      design_data=DESIGN,
      holdout_data=None, # inaccessible during search
   )

   update_research_memory(state, observation)
   state.cycle += 1

champion = freeze_lowest_loss_candidate(state.results)
state.holdout_locked = False

holdout_result = evaluate_once(champion, HOLDOUT)
benchmark_result = evaluate_once(historical_rate, HOLDOUT)

if (
   holdout_result.decision_loss < benchmark_result.decision_loss
   and holdout_result.brier < benchmark_result.brier
   and holdout_result.roc_auc > 0.50
):
   status = "VALIDATED FOR FURTHER CONTROLLED TESTING"
else:
   status = "REJECTED - RESEARCH ONLY"

export_manifest(state, champion, holdout_result, status)
```

B Reproducibility and Audit Checklist

Control	Implemented	Evidence
Explicit estimand	Yes	Ten-day path-dependent breach label
No future features	Yes	Features computed at t; labels use $t+1,\ldots,t+H$
Time-aware validation	Yes	Five expanding walk-forward folds

Control	Implemented	Evidence
Leakage gap	Yes	Gap equals $H = 10$ observations
Classical benchmark	Yes	Historical rate plus three classical learners
Quantum hypothesis can lose	Yes	Candidate classes include classical-only models
Candidate budget	Yes	Maximum 80; actual 47
Cycle budget	Yes	Maximum and actual five
Protected holdout	Yes	State lock through model selection
Champion frozen before test	Yes	Specification fixed before one-shot scoring
Negative decision path	Yes	Rejected—Research Only
Data fingerprint	Yes	SHA-256-derived identifier in manifest
Audit log	Yes	Timestamp, cycle, phase, event and details
Stress layer	Yes	Volatility, band, credibility and combined shocks
Production suitability	No	Synthetic educational laboratory only

C A Production Research Contract

Before an institutional run, the owner should sign off on the following contract:

1. **Purpose**: specify the decision supported and the exposure measured.
2. **Data**: identify sources, timestamps, revisions, entitlements and retention.
3. **Candidate space**: freeze admissible economic, quantum-inspired and ML families.
4. **Loss**: derive weights or utility from the business decision.
5. **Validation**: appoint a holdout custodian and independent challenge function.
6. **Tools**: enumerate local functions, MCP tools, APIs, plugins and credentials.
7. **Budgets**: cap cycles, candidates, compute, network calls and human escalations.
8. **Release gate**: define minimum performance, robustness and documentation.
9. **Monitoring**: define drift, breach, recalibration and retirement thresholds.
10. **Accountability**: name the human owner authorized to accept residual risk.

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